

51. Blondel, M., Teboul, O., Berthet, Q. & Djolonga, J. Fast Differentiable Sorting and Ranking. Preprint at <https://doi.org/10.48550/arXiv.2002.08871> (2020).

Supplementary information

Unsupervised Discovery of Functional Motifs from Protein Language Model with MotifAE

Figure S1-3

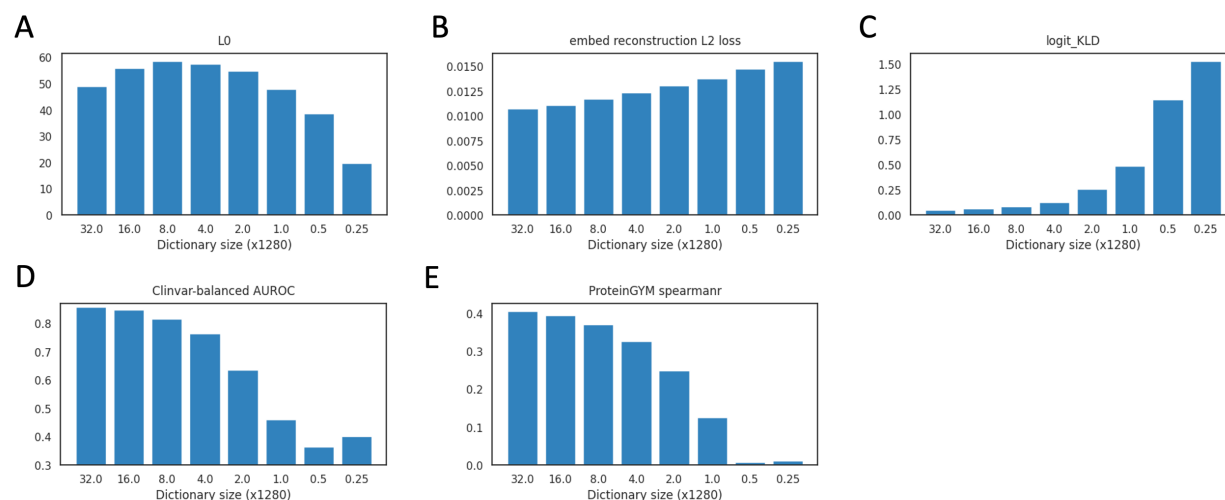


Figure S1. Comparison of different latent feature dimensions in MotifAE.

The values on the x-axis represent the expansion ratio relative to the embedding dimension (1,280). We note that the results shown here are obtained from MotifAE trained on normalized ESM2 embeddings in our preliminary experiment, whereas the results in the main figures are based on embeddings without normalization. Therefore, the absolute values in some figures may differ, but the overall conclusions should remain consistent. **(A)** Average number of non-zero latent feature activations per residue on the evaluation set. **(B)** Reconstruction loss on the evaluation set. **(C)** One residue in each evaluation protein was masked; amino acid probabilities were predicted using the MotifAE reconstruction with the fixed ESM2 MLP head and compared with predictions from ESM2 raw embeddings. A KLD value of zero indicates identical distributions. **(D-E)** Performance on mutation prediction using the wild-type log-likelihood ratio as the predictor. Mean Spearman correlation on ProteinGYM dataset was reported.